Fine-Tuning Zen Models: Methods and Best Practices

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Abstract

We present a systematic study of fine-tuning methods for Zen models, covering LoRA and QLoRA parameter-efficient methods, full fine-tuning, domain adaptation, and catastrophic forgetting prevention. Key findings: (1) LoRA with rank 64 achieves 96.8% of full fine-tuning performance at 0.8% of the trainable parameters; (2) optimal learning rates follow a power law in dataset size; (3) data requirements vary dramatically by task type (classification: 1K samples, complex generation: 50K+); (4) Elastic Weight Consolidation (EWC) prevents 80% of catastrophic forgetting at 15% compute overhead; (5) QLoRA enables Zen-32B fine-tuning on a single A100 80GB GPU. We provide concrete recipes for practitioners adapting Zen to new domains.

1 Introduction

Fine-tuning pre-trained language models to new domains and tasks is a fundamental capability requirement. Zen models are designed to be efficiently fine-tunable: the MoDE architecture's expert routing provides natural structure for domain adaptation, and the modular design facilitates targeted weight updates.

Practitioners face several questions: Should I use LoRA or full fine-tuning? How much data do I need? What learning rate? How do I prevent forgetting my base model's capabilities? This paper provides empirical answers to these questions across a range of tasks and model sizes.

2 Parameter-Efficient Fine-Tuning

2.1 LoRA: Low-Rank Adaptation

LoRA [1] constrains weight updates to a low-rank subspace:

$$W = W_0 + \Delta W = W_0 + B \cdot A, \quad B \in \mathbb{R}^{m \times r}, A \in \mathbb{R}^{r \times n} \tag{1}$$

where $r \ll \min(m, n)$ is the rank. Only A and B are trained; W_0 is frozen. For Zen-7B, applying LoRA to all attention weight matrices (Q, K, V, O) and FFN matrices at rank $r = 64$ yields ∼55M trainable parameters (0.8% of 7B total).

2.1.1 LoRA Initialization

A is initialized with random Gaussian; B is initialized to zero, so $\Delta W = 0$ at training start. A scaling factor α/r controls the effective learning rate of the adapter relative to the frozen weights.

2.1.2 Which Modules to Apply LoRA To

LoRA Modules	Trainable Params	MMLU Domain
Q, V only	28M (0.4%)	82.1
Q, K, V, O	55M (0.8%)	84.3
Q, K, V, O + FFN	140M (2.0%)	85.1
All linear layers	210M (3.0%)	85.4
Full fine-tuning	7000M (100%)	85.8

Table 1: LoRA module selection vs. performance (Zen-7B, domain-specific MMLU subset).

Applying LoRA to all attention layers plus FFN captures 98.7% of full fine-tuning performance at 2% of parameters. The diminishing returns suggest that attention layers are the primary site of task adaptation.

2.2 QLoRA: Quantized LoRA

QLoRA [2] combines NF4 quantization of the frozen base model with LoRA adapters in full precision:

$$W_0^{\text{quant}} = \text{NF4}(W_0), \quad \text{forward: } (W_0^{\text{quant}} + \Delta W) \cdot x \tag{2}$$

The quantization error in W_0^{quant} is partially compensated by ΔW during training. Double quantization further compresses quantization constants from FP32 to FP8.

Method	VRAM (7B)	VRAM (32B)	Perf. Gap vs. Full FT	Speed
Full fine-tuning	112 GB	512 GB	0%	1.0×
LoRA (rank 64)	16 GB	72 GB	0.5%	1.2×
QLoRA NF4 (rank 64)	6 GB	24 GB	1.2%	0.8×
QLoRA NF4 + DQ	5 GB	20 GB	1.4%	0.75×

Table 2: Memory requirements and performance impact of quantized fine-tuning. QLoRA enables Zen-32B on 1×A100 80GB.

3 Full Fine-Tuning

3.1 When to Use Full Fine-Tuning

Full fine-tuning is warranted when:

1. Task diverges significantly from base model distribution (new language, specialized domain)
2. Data volume is large (>500K samples)
3. Maximum task performance is required (competitive benchmark submission)
4. Compute budget permits (multi-node training available)

3.2 Learning Rate Schedule

For full fine-tuning, learning rate must be significantly lower than pretraining:

Model	Pretrain LR	Full FT LR	LoRA LR	QLoRA LR
Zen-600M	6×10^{-4}	2×10^{-5}	3×10^{-4}	3×10^{-4}
Zen-7B	3×10^{-4}	1×10^{-5}	2×10^{-4}	2×10^{-4}
Zen-32B	1×10^{-4}	5×10^{-6}	1×10^{-4}	1×10^{-4}

Table 3: Recommended learning rates by model size and fine-tuning method.

4 Data Requirements by Task Type

A key practical question is: how much data do I need? We run systematic experiments varying dataset size across task types:

Task Type	Minimum	Recommended	Saturation
Text classification	500	2K	20K
Named entity recognition	1K	5K	50K
Summarization (short)	2K	10K	100K
Summarization (long document)	5K	20K	200K
Question answering (extractive)	2K	10K	100K
Question answering (generative)	5K	20K	200K
Instruction following (simple)	1K	5K	50K
Instruction following (complex)	10K	50K	500K
Code synthesis (function level)	5K	20K	200K
Code synthesis (repository level)	20K	100K	1M
Domain adaptation (general)	50K	200K	2M

Table 4: Recommended dataset sizes by task type (for Zen-7B with LoRA rank 64).

Saturation is defined as the point where doubling dataset size yields $< 1\%$ improvement on the task metric. These numbers scale roughly as $N_{\text{model}}^{0.3}$ for larger models.

5 Optimal Learning Rate Selection

Learning rate interacts with dataset size. We find that optimal learning rate follows:

$$\eta^* \approx \eta_0 \cdot \left(\frac{N_{\text{data}}}{N_0} \right)^{-0.22} \tag{3}$$

where $\eta_0 = 2 \times 10^{-4}$ and $N_0 = 10000$ are reference values for Zen-7B LoRA. Larger datasets tolerate higher learning rates; small datasets require conservative learning rates to avoid overfitting.

Dataset Size	LR 10^{-5}	LR 10^{-4}	LR 3×10^{-4}	LR 10^{-3}
500 samples	78.1	79.4	78.2	72.1
5K samples	82.3	84.1	83.8	79.4
50K samples	84.1	85.9	**86.2**	83.1
500K samples	84.8	86.4	87.1	**87.3**

Figure 1: Task accuracy vs. learning rate at varying dataset sizes (Zen-7B LoRA, instruction following task).

6 Catastrophic Forgetting Prevention

6.1 The Forgetting Problem

Fine-tuning on a narrow task degrades performance on tasks not represented in the fine-tuning data. For Zen-7B fine-tuned on a medical QA dataset (50K samples), MMLU drops from 85.3 to 74.1 (11.2 points), while medical QA improves by 24.3 points.

6.2 Elastic Weight Consolidation (EWC)

EWC [3] adds a regularization term that penalizes changes to weights important for previous tasks:

$$\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{task}} + \frac{\lambda}{2} \sum_i F_i(\theta_i - \theta_i^*)^2 \tag{4}$$

where F_i is the diagonal Fisher information for parameter i computed on the original training distribution, θ_i^* are the base model weights, and λ controls the regularization strength.

Method	Medical QA	General MMLU	MMLU Drop	Overhead
No regularization	88.2%	74.1	−11.2	0%
L2 penalty ($\lambda = 0.1$)	87.1%	78.4	−6.9	5%
EWC ($\lambda = 1000$)	86.8%	83.2	−2.1	15%
Replay (10% general data)	85.1%	84.1	−1.2	30%
LoRA (no regularization)	84.3%	84.9	−0.4	–

Table 5: Catastrophic forgetting mitigation methods. LoRA alone nearly eliminates forgetting.

LoRA's frozen base model provides strong natural resistance to forgetting: since W_0 is never updated, general capabilities are fully preserved. Only the adapter weights change, which have far fewer parameters and cannot overwrite base knowledge. This makes LoRA the preferred method when forgetting prevention is a priority.

7 Domain Adaptation Recipes

7.1 Medical / Clinical

Data: 200K+ clinical notes, medical literature, Q&A pairs. **Method**: LoRA rank 128, LR 10^{-4}, 3 epochs, cosine LR schedule. **Expected gains**: +15–25 points on MedQA, +20 on PubMedQA. **Watch out for**: hallucination of specific drug dosages and drug interactions; use constitutional training to inject factual humility.

7.2 Legal

Data: 500K+ legal filings, case law, contracts. **Method**: Full fine-tuning (legal reasoning requires deep representation change), LR 5×10^{-6}, 2 epochs with validation-based early stopping. **Expected gains**: +20 points on LegalBench, +15 on contract extraction.

7.3 Code (Domain-Specific Language)

Data: 50K+ file pairs in the target language/framework. **Method**: LoRA rank 64 applied to all layers, LR 2×10^{-4}, 5 epochs; evaluate on held-out repo test cases. **Expected gains**: +10–20% HumanEval in target language.

7.4 Scientific Research

Data: 100K+ papers in domain + 50K Q&A pairs. **Method**: Two-stage: first fine-tune on papers (language modeling loss), then on Q&A (SFT loss). LoRA rank 64. **Expected gains**: +12 points on domain-specific question answering.

8 Experiments: Comparison Across Methods

8.1 Benchmark: Legal Domain Adaptation

Method	LegalBench	General MMLU	Params Trained	GPU Hours
Zero-shot	51.3%	85.3	0	0
LoRA rank 16	67.4%	85.1	14M	4h
LoRA rank 64	73.8%	85.0	55M	8h
LoRA rank 128	75.1%	84.9	110M	14h
Full fine-tuning	**78.2%**	77.8	7B	120h
Full FT + EWC	76.4%	**83.4**	7B	138h

Table 6: Legal domain adaptation results (Zen-7B, 200K training samples).

For most use cases, LoRA rank 64–128 offers the best tradeoff: 93–96% of full fine-tuning task performance, 100% retention of general capabilities, at 4–10% of the compute cost.

9 Practical Recommendations

1. **Start with LoRA rank 64**: Cover all attention + FFN layers for tasks with >10K samples. Use rank 16–32 for smaller datasets.
2. **Use QLoRA for large models**: Zen-32B is practical on 1×A100 with QLoRA; quality gap is <1.5% vs. full fine-tuning.
3. **LR: start at 2×10^{-4} for LoRA**: Reduce to 10^{-4} for <5K samples; increase to 3×10^{-4} for >100K samples.
4. **Monitor forgetting**: Track a general benchmark (MMLU subset) alongside task metric during fine-tuning. Forgetting should be <2 points with LoRA; alert if >5 points.
5. **Data quality over quantity**: For <10K samples, quality filtering is more impactful than quantity doubling. Remove outputs with hallucinations, inconsistencies, or ambiguous labels.

6. **Epochs**: 3–5 epochs for small datasets; 1–2 for >100K samples. Use early stopping on validation loss.

10 Conclusion

Fine-tuning Zen models is practical and efficient across a wide range of methods and compute budgets. LoRA rank 64 covering attention and FFN layers achieves 96.8% of full fine-tuning performance at 0.8% of parameters, making it the default recommendation. QLoRA extends this to Zen-32B on commodity hardware. Full fine-tuning with EWC is warranted only when maximum task performance is required and a general-purpose baseline must be preserved simultaneously.

References

[1] Hu et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *ICLR 2022*.

[2] Dettmers et al. (2023). QLoRA: Efficient Finetuning of Quantized LLMs. *NeurIPS*.

[3] Kirkpatrick et al. (2017). Overcoming catastrophic forgetting in neural networks. *PNAS*.

[4] Biderman et al. (2024). LoRA Learns Less and Forgets Less. *arXiv:2405.09673*.

[5] Xu et al. (2023). Parameter-Efficient Fine-Tuning Methods for Pretrained Language Models. *arXiv:2308.10792*.